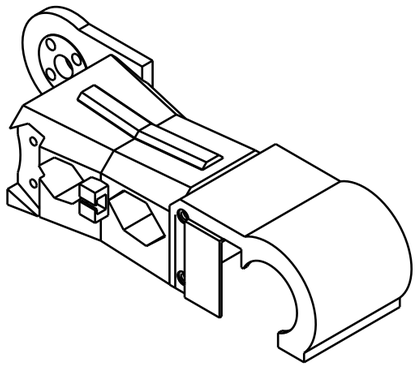
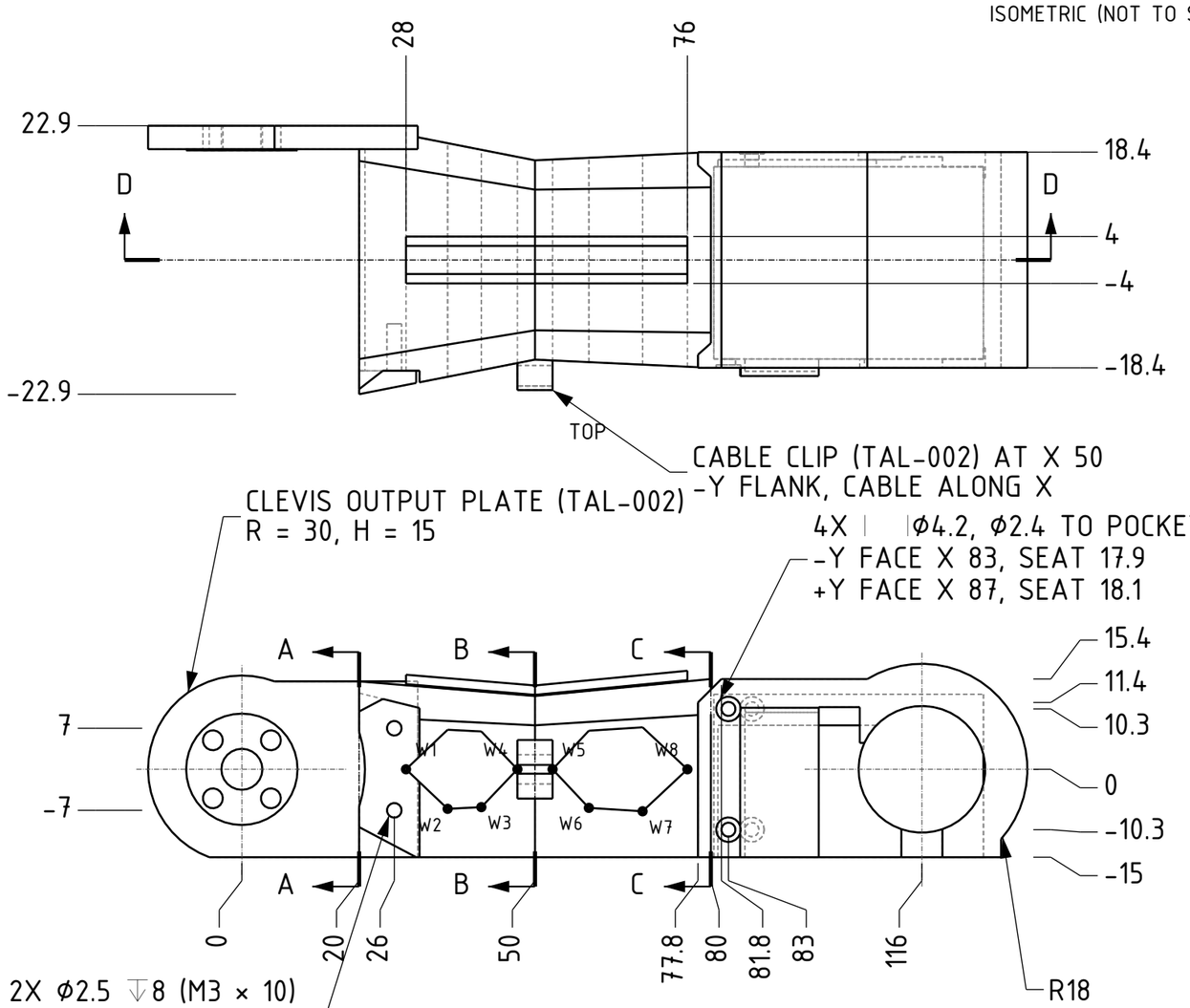


ISOMETRIC (NOT TO SCALE)



WINDOW POINTS (X, Z), THROUGH Y

PT	X	Z
W1	28	0
W2	35.1	±6.75
W3	40.85	±6.46
W4	47	0
W5	53	0
W6	59.19	±6.58
W7	68.35	±7.16
W8	76	0

FEATURES (in order)

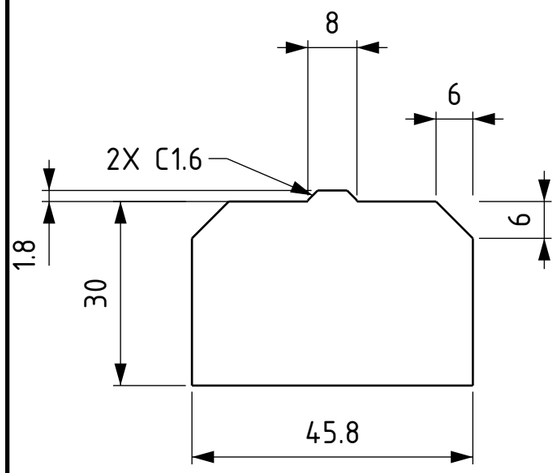
1. Clevis output plate only (TAL-002, +Y plate with its horn pad), R = 30, H = 15. The idler plate is the separate part TAL-110.
2. Beam: ruled loft of the profiles A, B, C (sheet 2) at x 20, 50, 80. Every profile sits on z −15, so the underside is flat; the waist dips on top only.
3. Servo shell: XZ polygon (77.8, −15) (116, −15) (116, 15.4) (81.8, 15.4) (77.8, 11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 116.
4. Flat underside: cut everything below z −15 (trims the clevis hub and the R18 cylinder).
5. Nose floor: box x 125 to 129.5, y ±18.4, z −15 to −11; ties the shell walls together on the bed (the servo entry cut later trims its back to x 126.5).
6. Doubler on the −Y wall: box x 85 to 99.6, y −19.8 to −18.4, z −15 to 10.5 (leaves 0.8 over the bump channel; out of the next link's reach).
7. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 76.
8. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
9. Cable clip (TAL-002) on the −Y flank at x 50, cable along X.
10. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes; the output pad stays.
11. Pocket P (sheet TAL-001) with its servo frame origin P0 at (116, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along −Z.
12. Servo entry from below (pocket P run on down, TAL-001; horns off): case box x 80.5 to 126.5, y −16.9 to 15.9, z below 12.8 (opens the floor under the whole servo). +Y wall: rib channel x 81.2 to 112.5, y 15.5 to 17.1, z below 7.5; spline slot x 116 ±3.5, y to 17.6, z below 0. −Y wall: bump channel x 85.8 to 98.8, y −19 to −16.5, z below 9.7; plug window x 98.4 to 105.4, z below 10.6; notch through the wall x 98.4 to 119.5, z below 0 (joins the plug window and the idler hole).
13. Case screw holes (TAL-002): +Y face (seat 18.1, over the rib channel; heads stand 0.7 proud) at x 87, z ±10.3; −Y face (seat 17.9) at x 83, z ±10.3.
14. Idler plate pocket: the TAL-110 outline (circle R10.5 + neck N1 to N4) grown 0.3, cut from y −23.9 to −18.9.
15. Idler plate screw holes: 2X ⌀2.5 at x 26, z ±7, from y −18.9 8 deep (M3 × 10).

NOTES

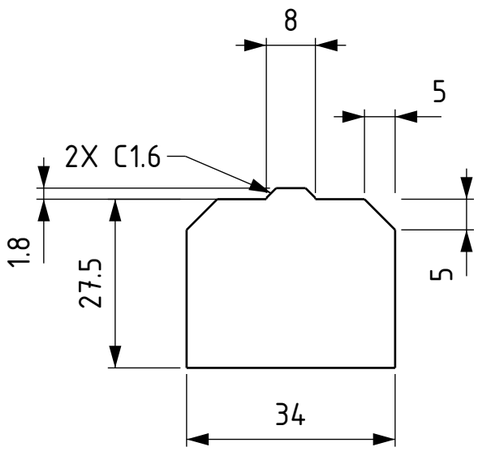
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/−0.2 unless noted; servo pocket clearances are already included.
- Fit the elbow servo from below, shaft to +Y, with both horns off; then the case screws (2X M2 × 6 on +Y, 2X M2 × 4 on −Y) and refit the horns.
- The elbow servo's bus plugs come out through the −Y shell wall.
- Print: flat underside down; no support (servo-opening roof and window tops bridge). The flat underside is z −15.

THIRD ANGLE PROJECTION

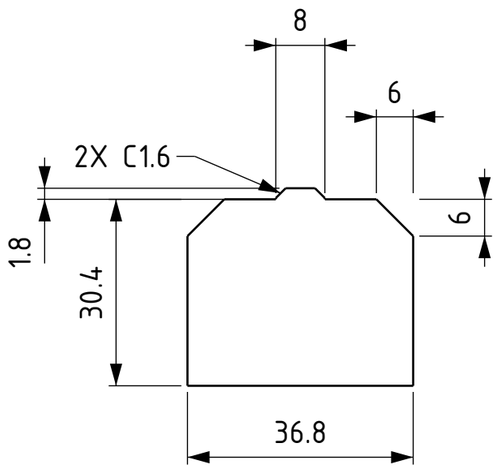
WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 48 g
PART Upper arm (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 10 of 21



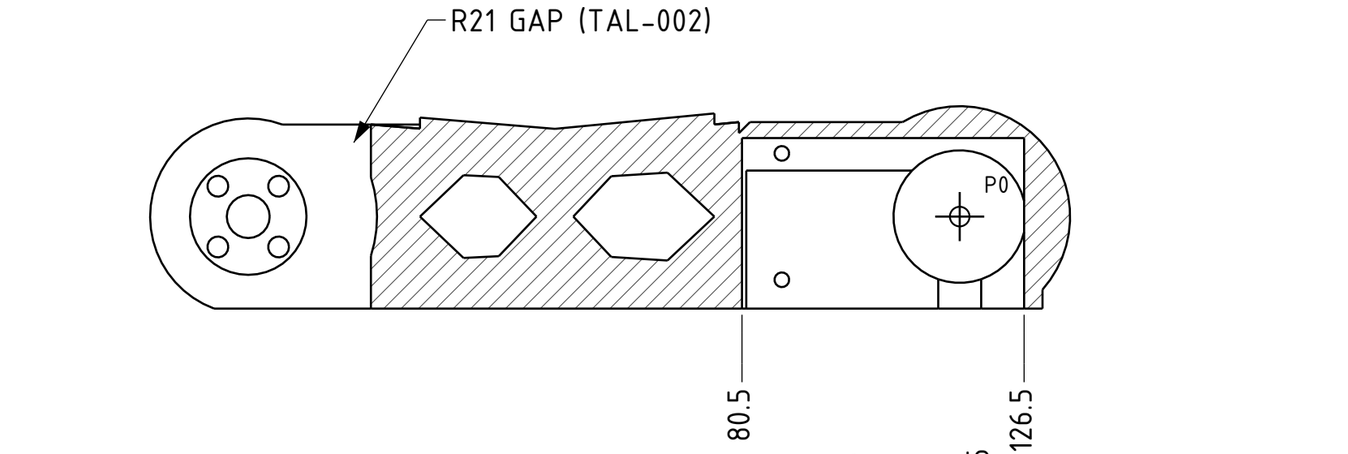
PROFILE A–A (X 20)



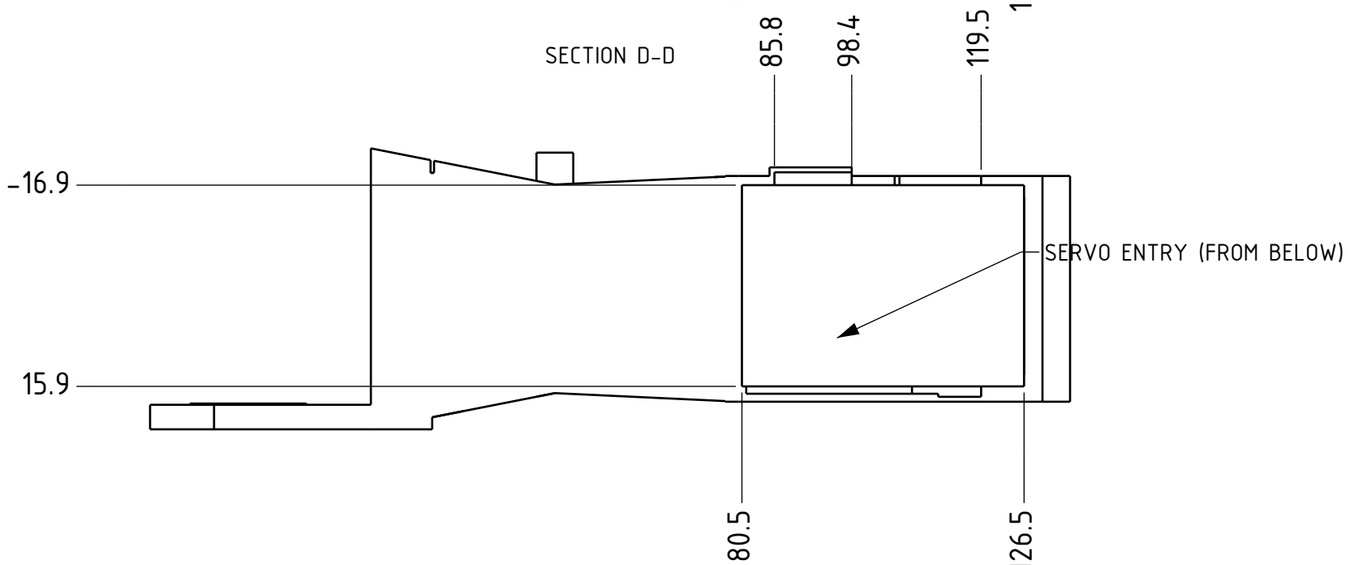
PROFILE B–B (X 50)



PROFILE C–C (X 80)



SECTION D–D



BOTTOM

PROFILES AND SECTION

Profiles A, B, C: beam section (top corners chamfered, underside flat and square-cornered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1. Each sits on z −15 (flat underside) with its top at the height given.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

Section D–D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS –
PART Upper arm (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 11 of 21